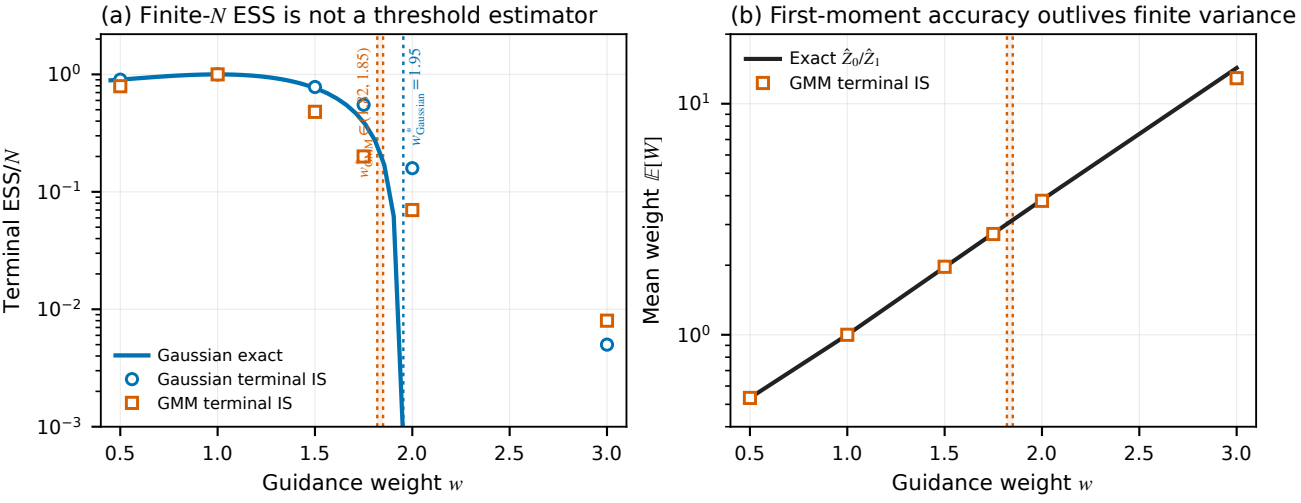
(a) Finite-N ESS is not a threshold estimator
Terminal ESS/N
10^0
10^{-1}
10^{-2}
10^{-3}
$w_{\mathrm{GMM}} \in (1.82, 1.85)$
$w^*_{\mathrm{Gaussian}} = 1.95$
Gaussian exact
Gaussian terminal IS
GMM terminal IS
0.5
1.0
1.5
2.0
2.5
3.0
Guidance weight w
(b) First-moment accuracy outlives finite variance
Mean weight $\mathbb{E}[W]$
10^1
10^0
Exact $\hat{Z}_0/\hat{Z}_1$
GMM terminal IS
0.5
1.0
1.5
2.0
2.5
3.0
Guidance weight w